وزارة الاتـــــــــصالات
وتكنولوجيا المعلومات

الرواد الرقميون
DIGILIANS

Project Title

GlucoGuide

AI AI-Based Diabetes Prediction, Classification & Personalized Recommendation System

From Prediction to Prevention

Track: Applied Ai & Data Analytics

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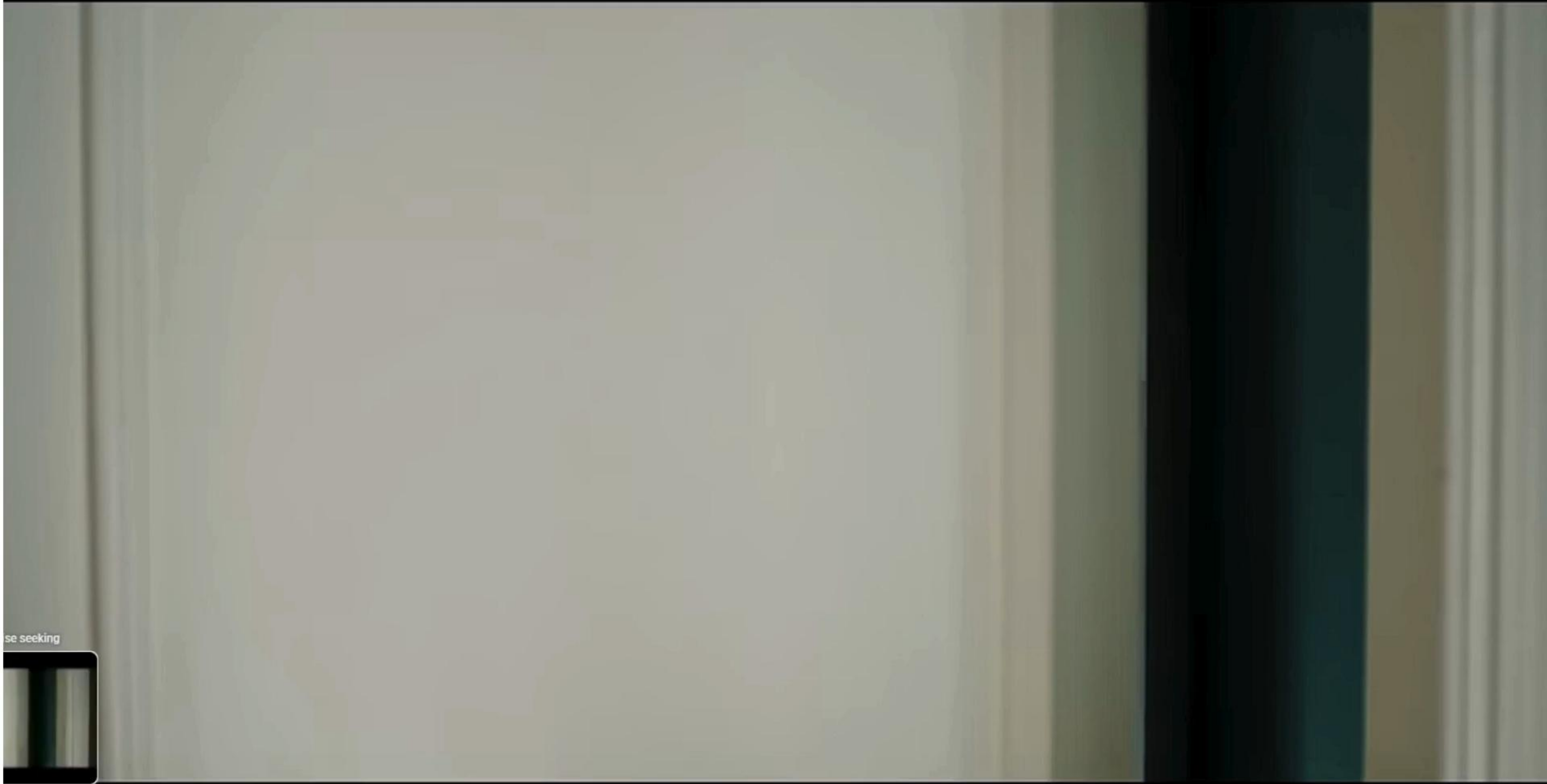
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CONTENTS

Agenda

• LIVE DIABETES INTELLIGENCE • 2024–2025

From Prediction to Prevention.

A modern analytics view of the global and Egyptian diabetes landscape — every figure, chart, and insight preserved exactly as reported by WHO and the IDF Diabetes Atlas.

Explore the data → | Prevention cloud ↗

589M
adults living with diabetes worldwide, 2024

13.2M
adults living with diabetes in Egypt, 2024

1/9s
a person dies from diabetes somewhere in the world

01 — The Global Picture

How diabetes has grown worldwide — and where it's heading. All figures from WHO (Nov 2024) and the IDF Diabetes Atlas, 11th edition (2025).

KPI 01
589M
adults with diabetes — 1 in 9 people worldwide

KPI 02
853M
projected by 2050 — 1 in 8 adults

KPI 03
3.4M
deaths attributed to diabetes in 2024

KPI 04
$1.02T
spent globally on diabetes care — up 338% in 17 years

KPI 05
43%
of adults with diabetes (252M) don't know it yet

One platform, end to end

Global Platform

DASHBOARD

- Global Platform
- Patients
- Egypt Spotlight
- Data You Know
- Prevention & Control
- Resources

Search countries, regions, indicators...

Global Platform

KPI 01

589M

adults with diabetes — **1 in 8** people worldwide

KPI 02

853M

prediabetes in 2000 — **1 in 6** adults

KPI 03

3.4M

deaths attributed to diabetes in 2021

KPI 04

$1.02T

global health cost of diabetes care — up **331%** (17 years)

KPI 05

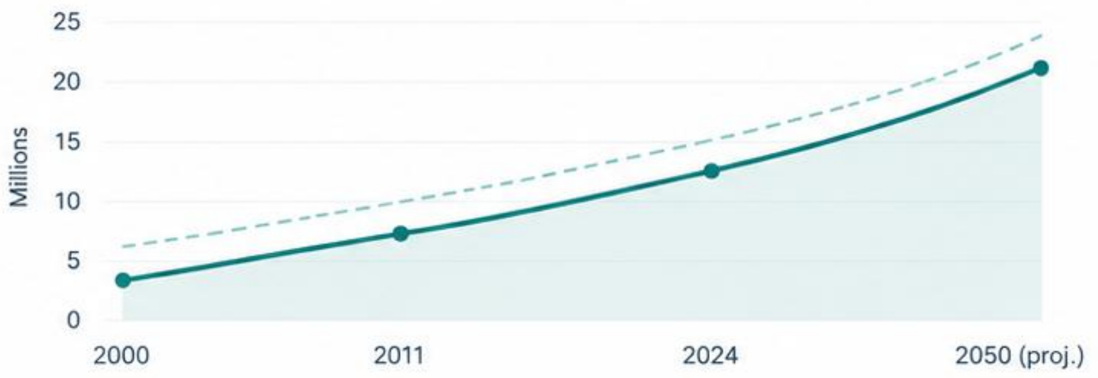
43%

of adults with diabetes (12.1M) don't know they have it

Global diabetes growth, 2000 – 2050

TREND

Adults with diabetes (millions)



Diabetes (millions)   Projected trend

Prevalence by region

REGION

Age-standardised, 2021

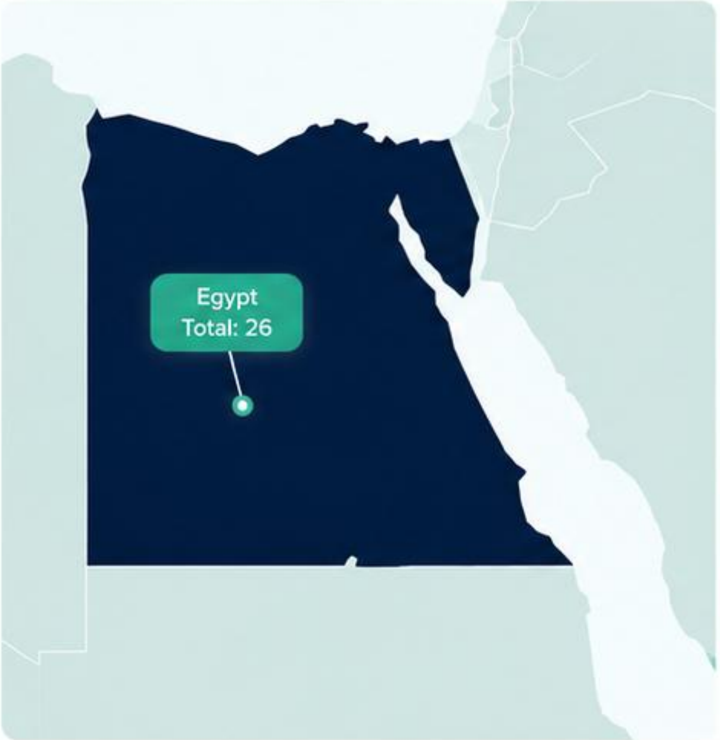
The Western Pacific region has the highest prevalence of diabetes **(17.5%)**.

Country & Regional Insights

Egypt's diabetes burden and its position in the region

Egypt

Diabetes country report 2000 – 2050



People with diabetes

13.2M

(in 1,000s)

Age-standardised prevalence

22.4%

(vs. global avg. ~2x)

Highest in the region

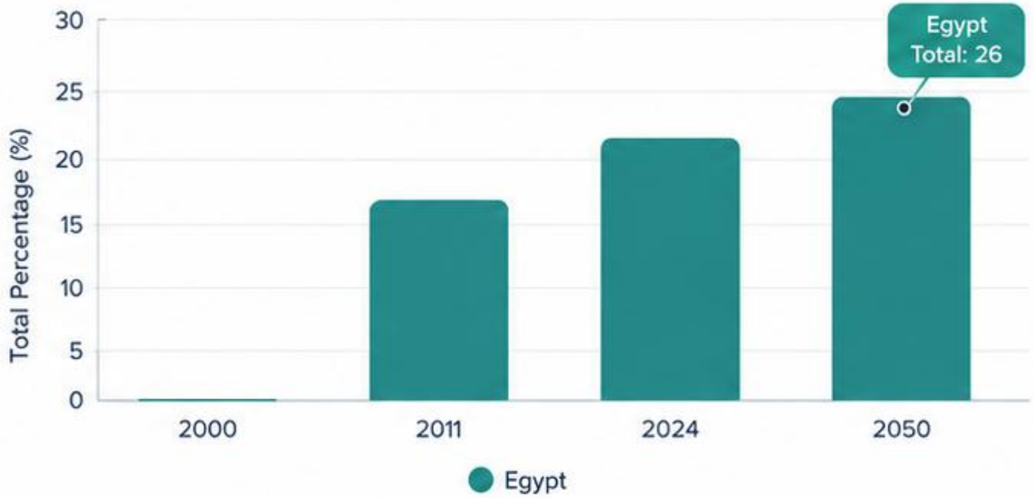
#1

in MENA

Age–standardised prevalence of diabetes, %

Compare countries:

Egypt | Select a country | Select a country



Egypt has the highest diabetes burden in the MENA region, with a 26% age-standardised prevalence in 2050.

Diabetes: A Growing Global Health Challenge

A chronic metabolic disorder

Diabetes Mellitus causes abnormal blood glucose levels through insufficient insulin production or ineffective insulin use.

Serious complications

Cardiovascular disease, kidney failure, nerve damage and vision impairment follow late detection.

Late diagnosis is the norm

Traditional diagnosis relies on lab tests and clinical evaluation performed only after symptoms appear.

Why Machine Learning?

ML can analyse large healthcare datasets and reveal hidden patterns of diabetes risk before symptoms appear.

Clinical & lifestyle factors used in this project:

- Glucose
- HbA1c
- Blood pressure
- BMI
- Cholesterol
- Smoking
- Physical activity
- Sleep duration
- Family history

Problem Statement

Prediction without explanation

Systems optimise accuracy but give no rationale, so users and clinicians cannot trust the output.

No personalized guidance

Few systems translate a prediction into actionable, individual preventive advice.

No lifestyle simulation

Users cannot explore how changing smoking, activity, sleep or BMI would affect their risk.

Limited practical deployment

Many models stay inside notebooks and never reach an interactive, user-friendly application.

Fragmented feature use

Clinical and behavioural features are rarely combined in one unified predictive framework.

Motivation & Research Gap

EXISTING APPROACHES

- High predictive accuracy reported
- Explanations often technical or absent
- Little individualized preventive guidance
- No counterfactual lifestyle analysis
- Offline notebooks, no real deployment
- Clinical OR lifestyle features, rarely both

VS

PROPOSED SYSTEM

- Machine Learning prediction (5 algorithms)
- SHAP-based explainability
- Personalized preventive recommendations
- Lifestyle simulation of risk factors
- Grounded RAG nutrition assistant
- Interactive Streamlit web application

Our system unifies accuracy, explainability, personalization, and real-world usability
all components integrated within a **single unified platform**.

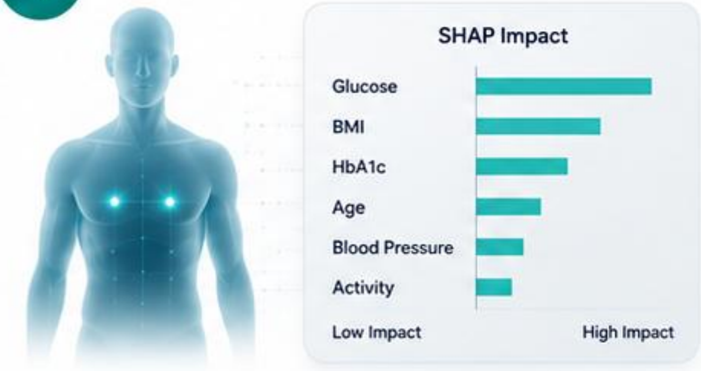
Project Objectives

1 Data Processing & Analysis

2 Machine Learning

3 Evaluation & Optimization

4 Explainable AI



5 Personalized Recommendations

6 Interactive Deployment

Meet GlucoGuide

From preaiction to prevention

A Growing Need for
Smarter Diabetes Care

A large diabetes burden is creating demand for continuous, personalized and **AI-enabled** care.

GlucoGuide
Predict. Explain. Personalize. Prevent.

01 GLOBAL NEED

589M
Adults living with diabetes worldwide

252M
Estimated to be unaware of their condition

Source: IDF Diabetes Atlas 2024

02 DIGITAL MARKET

$20.2B
Global digital diabetes management market – 2025

13.3%
CAGR projected growth through 2035

Source: Grand View Research, 2024

03 EGYPT OPPORTUNITY

13.2M
Adults digith diabetes in Egypt

62%
Estimated proportion undiagnosed

Source: IDF Diabetes Atlas 2024

THE OPPORTUNITY FLOW

Growing Diabetes Burden → More Patient Data → AI & Digital Health Adoption → Market Growth & Investment → GlucoGuide

GlucoGuide
Predict → Explain
Personalize → Prevent

KEY TAKEAWAY

The opportunity is shifting from simply managing diabetes to **predicting risk and enabling personalized prevention.**

Huge unmet need + strong market growth + AI innovation = the perfect time for **GlucoGuide** to make an impact.

GlucoGuide
Predict. Explain. Personalize. Prevent.

One platform, end to end

A single connected pipeline: from raw patient data to a **risk score**, a clear **explanation**, grounded **nutrition guidance**, and **clinical oversight**.

Patient Data
Collect and consolidate patient information

AI Risk Prediction
Advanced AI models assess risk accurately

SHAP Explanation
Explainable insights you can trust

RAG Nutrition Support
Personalized nutrition guidance powered by evidence

Web Interface
Intuitive, secure platform for patients and clinicians

Doctor Monitoring
Monitor, review, and make better decisions

Better decisions. | Healthier outcomes. | Lower costs. | Stronger prevention.

GlucoGuide connects AI, clinical expertise, and nutrition science to deliver proactive diabetes care.

Target users & use cases

GlucoGuide
Predict. Explain. Personalize. Prevent.

GlucoGuide turns data into **actionable insight** — moving diabetes care from reactive management to **proactive prevention**.

Patients

Enter data or upload lab report
Easily input or upload your data

Receive risk score & explanation
Understand your risk clearly

Ask nutrition questions (RAG)
Get reliable, personalized answers

Track results over time
Monitor your progress

GlucoGuide

From Prediction to Prevention to Value

AI PREDICTION
Accurate risk assessment

EXPLAINABILITY
Understand the "why" behind the prediction

PERSONALIZATION
Tailored recommendations that fit each individual

PREVENTION
Lifestyle changes today, healthier tomorrow

Doctors & Clinicians

Monitor multiple patients
View and follow patient trends

Review high-risk cases first
Prioritize what matters most

Verify OCR-extracted values
Ensure accuracy with confidence

Use SHAP for clinical discussion
Explainable insights you can trust

89%
Patients want personalized recommendations*

70%
Chronic disease cost driven by preventable complications*

$1.5T+
Potential annual savings with effective diabetes prevention*

Better outcomes. Lower costs. Healthier futures.

Sources: IDF Diabetes Atlas 2024, Grand View Research (2024), McKinsey (2023)
*Figures from industry reports and studies

BUSINESS MODEL CANVAS

A clear view of how we create and deliver value.

Key Partners

- Hospitals & Clinics
- Nutrition Specialists
- Medical Labs (OCR)
- Cloud / API Providers

Key Activities

- Model Training & Tuning
- RAG Knowledge Curation
- Platform Maintenance
- SHAP Explainability

Value Propositions

- Early, explainable risk detection
- Evidence-grounded nutrition Q&A
- Faster, less manual data entry
- Doctor–patient shared trust

Customer Relationships

- Self-service Portal
- Clinician Review
- Transparent AI Explanations

Customer Segments

- At-risk Individuals & Patients
- Doctors & Clinicians
- Clinics & Health Platforms

Key Resources

- Clinical Dataset (100K)
- Trained XGBoost Model
- Nutrition Reference Corpus

Channels

- Web Application
- Doctor Referral
- Clinic Onboarding

Cost Structure

- Cloud Hosting & Compute
- Model Retraining
- Data & Content Licensing

Revenue Streams

- Clinic / Institution Licensing
- Subscription Tiers
- API Access for Partners

From Diabetes Management to Intelligent Prevention

Most platforms help you **manage** diabetes. GlucoGuide goes further — from **prediction** to **personalization**.

Key Capabilities	mySugr Tracking & Logbook	SugarBuddy AI Diabetes Companion	DARIO Digital Therapeutics	GlucoGuide Prediction · Explain · Prevent
Glucose Tracking	✓	✓	✓	✓
Lifestyle Insights	✓	✓	✓	✓
Risk Prediction	—	◐	◐	✓
Explainable AI / SHAP	—	—	—	✓
Personalized Recommendations	◐	✓	✓	✓
Lifestyle Simulation	—	—	—	✓
Nutrition AI / RAG	—	—	—	✓
Interactive Platform	✓	✓	✓	✓
	Strong tracking	AI insights	Care support	Unified AI Platform

GlucoGuide Advantage

Explainable AI
Know why, not just what

Personalized Recommendations
Advice tailored to you

Lifestyle Simulation
See how changes reduce your risk

Nutrition Assistant (RAG)
Evidence-based answers to your questions

Our Journey: Track › Predict › Explain › Personalize › Prevent

Machine learning pipeline

1 Data

Collect & consolidate raw data

2 Preprocessing

Clean, handle missing values, validate

3 Feature Engineering

Create meaningful features

4 Feature Selection

Select the most important features

5 Model Training

Train models on prepared data

6 Model Evaluation

Evaluate performance using key metrics

7 Final Model

Deploy the best performing model

Mutual Information Analysis

Identify and retain features with the strongeest relationships.

SMOTE Resampling Experiments

Address class imbalance and improve minority class representation.

PCA Dimensionality Experiment

Reduce dimensionality while preserving important information.

Five-Model Comparison

Compare multiple algorithms to select the best performing model.

Dataset & Data Understanding

100,000
Patient records

31
Original features

kaggle
Diabetes Healthcare Dataset

diagnosed_diabetes
Target variable

Demographic

- Age
- Gender
- Ethnicity
- Education level
- Income level
- Employment status

Lifestyle

- Smoking status
- Alcohol per week
- Physical activity min/week
- Diet score
- Sleep hours
- Screen time

Medical History

- Family history of diabetes
- Hypertension history
- Cardiovascular history

Clinical & Laboratory

- BMI
- Waist-to-hip ratio
- Blood pressure (systolic/diastolic)
- Heart rate
- HDL & LDL cholesterol
- Triglycerides
- Fasting & postprandial glucose
- Insulin
- HbA1c

Additional variables diabetes_risk_score and Diagnosis_encoded were excluded as direct target proxies (data-leakage prevention).

Data Preprocessing

1. Raw Data 100,000 × 31 (100K+)
2. Cleaning & Validation
3. Exploratory Data Analysis
4. Encoding & Scaling (0101 1010)
5. Feature Engineering
6. MI Feature Selection (MI)
7. Stratified Split 80 / 15 / 5
8. Class Imbalance Study
9. Model Training 5 algorithms
10. GridSearchCV + Stratified CV (CV)
11. Evaluation 5 metrics
12. Final Model XGBoost

Reported experimental setup

80,000 train / 15,000 validation / 5,000 test (stratified) · StandardScaler · One-Hot Encoding · engineered postprandial-to-fasting glucose ratio · low-MI features removed (alcohol/week, screen time, diet score, diastolic BP) · 32-feature matrix, 28 descriptors used for prediction.

Class imbalance

Original vs. ROS vs. SMOTE vs. SMOTEENN were compared — the Original (imbalanced) data gave the best overall balance (Acc 0.9169 · ROC-AUC 0.9510) and was used for final training.

Machine Learning Models & Experimental Setup

SVM
RBF kernel, cross-validated — captures nonlinear boundaries

Random Forest
Bootstrap-aggregated decision trees; stable, resistant to overfitting

XGBoost
Sequential gradient boosting; strong on structured healthcare data

LightGBM
Efficient boosting for large-scale tabular data, fast training

MLP
Neural benchmark for nonlinear feature interactions

Training & Evaluation Strategy

- Identical preprocessed data for every model
- GridSearchCV + stratified cross-validation
- Validation set for tuning, unseen test set for generalization
- MLflow logs parameters, metrics and runs

Evaluation Metrics

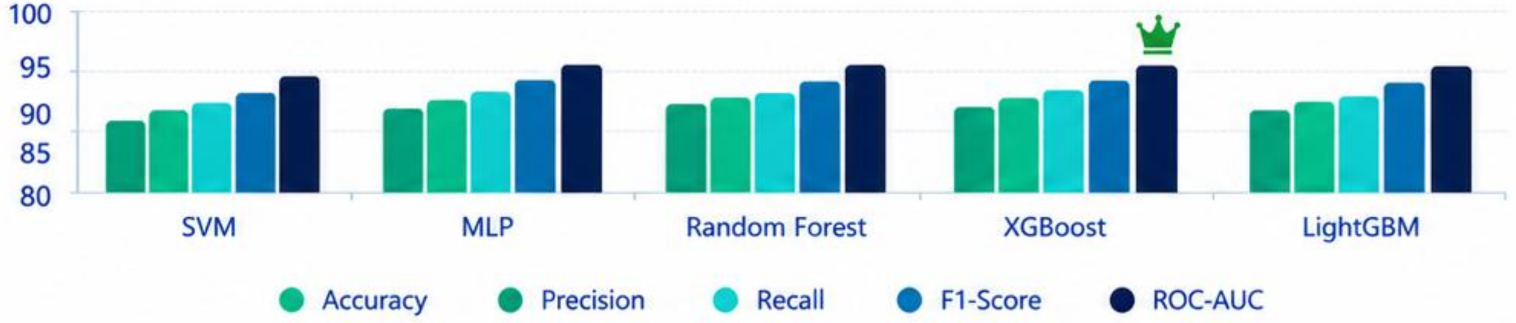
- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Baseline Results — Model Comparison

Validation performance of all five models on the full feature set (before optimization experiments).

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
SVM	86.93%	87.40%	86.93%	86.95%	93.88%
MLP	90.47%	91.23%	90.47%	90.42%	94.42%
Random Forest	91.63%	92.59%	91.63%	91.52%	94.86%
XGBoost	**91.63%**	**92.58%**	**91.63%**	**91.54%**	**95.12%**
LightGBM	91.20%	91.95%	91.20%	91.10%	94.95%

Model Performance Comparison



Key Takeaway

XGBoost achieved the highest performance across all metrics, with the best ROC-AUC score (95.12%).

RESULTS

Model Optimization

Original Features → Engineered Clinical Features → PCA / Dimensionality Analysis → Re-training → Re-evaluation

Engineered Features Tested

- ✓ BMI-to-Waist-Hip Ratio (central obesity)
- ✓ Total Cholesterol-to-HDL Ratio (cardiometabolic risk)
- ✓ Triglyceride-Glucose (TyG) Index (insulin resistance)
- ✓ Sleep-to-Screen Time Ratio (lifestyle balance)

Findings

- Engineered features were evaluated using Mutual Information (MI) and model-based analysis.
- The Postprandial-to-Fasting Glucose Ratio showed useful predictive information and was retained.
- PCA reduced the feature space from **32** features to **3** components while retaining **95.57%** of the variance.
- Models were re-trained and re-evaluated to determine the final configuration.

Optimized / Final Results (validation)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost	**91.61%**	**92.56%**	**91.61%**	**91.52%**	**95.14%**
MLP	90.22%	90.71%	90.22%	90.16%	94.55%
SVM	86.13%	86.19%	86.13%	86.09%	93.73%

All models were re-evaluated after optimization; all runs tracked in MLflow.

Neural Network vs. XGBoost

GlucoGuide
From Prediction to Prevention

Performance and computational efficiency comparison under the **same** feature-engineering setting (latest MLflow runs).

PREDICTIVE PERFORMANCE

Metric	Neural Network	XGBoost	Improvement
Accuracy (%)	91.51%	**91.61%**	+0.10 pp
F1-Score	91.42%	**91.52%**	+0.10 pp
ROC-AUC	95.11%	**95.14%**	+0.03 pp

XGBoost slightly outperformed the Neural Network across all predictive metrics.

COMPUTATIONAL EFFICIENCY

	Neural Network	XGBoost	
Training Time (seconds)	462.07 s	2.48 s	~186x Faster
Inference Time (seconds)	0.419 s	0.037 s	~11x Faster
Trainable Parameters	5,189	83,908	~16x More Parameters

Despite having substantially fewer parameters, the Neural Network did not achieve faster training or inference.

Why XGBoost?

- Slightly better performance
- Substantially faster execution
- SHAP-based explainability

→ **XGBoost was selected as the final model for the proposed diabetes prediction system.**

Final Model Selection — XGBoost

Accuracy	Precision	Recall	F1-Score	ROC-AUC
91.61%	92.56%	91.61%	91.52%	95.14%

XGBoost

Best Overall Performance

Why XGBoost was selected

- High Performance
- Robust & Stable
- Handles Complexity
- SHAP Compatible
- Efficient
- Easy Deployment

Explainable AI with SHAP

From predictions to understanding.

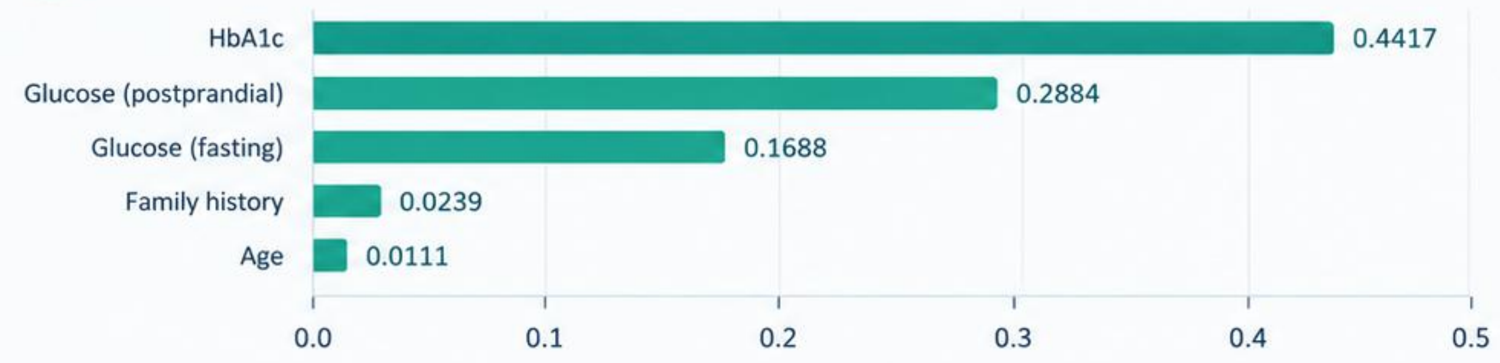
Prediction + **Feature Contributions** = **Explainable Prediction**

Global Importance
Ranks features by overall impact.

Individual Explanation
Shows how features influence each patient.

Transparency & Trust
Builds confidence in the model's decisions.

Top Predictors (SHAP)



Key Takeaways

- **HbA1c** is the strongest predictor.
- Glucose levels and family history also play a key role.
- **SHAP** provides clear, patient-level explanations.

Evidence-Grounded RAG Nutrition Assistant

Knowledge Source: Nutritional Management of Diabetes Mellitus — Frost (2003)

KNOWLEDGE PREPARATION

Document Cleaning
Remove noise and preserve structure

→

Section-Aware Chunking
Preserve topic context with controlled overlap

→

BM25 + Semantic Index
Build searchable index for efficient retrieval

QUESTION → ANSWER

User Question
Natural language query from the user

→

Hybrid Retrieval (50/50)
BM25 (lexical) + Semantic (meaning) to find relevant candidates

→

Cross-Encoder Reranking
Re-score and reorder candidates by true relevance to the question

→

Context Filtering
Remove weak & duplicate results and keep within context limit

→

Grounded Answer
Generate answer based only on retrieved evidence

271 Pages | **312** Chunks | High-quality, evidence-based nutrition recommendations

GlucoGuide in Action

A live web platform that delivers AI-powered diabetes prediction and personalized clinical insights.

Real-time Clinical Dashboard

Live monitoring of key health metrics, risk score, and confidence level.

Explainable AI Insights

Top factors and clear explanations that make predictions transparent.

AI Recommendations

Personalized, evidence-based suggestions to improve patient outcomes.

Interactive AI Assistant

Ask questions in natural language and get instant, reliable answers.

Diabetes Intelligence Platform

GlucoGuide

Helping individuals and healthcare professionals predict diabetes risk, classify diabetes stage, and support clinical decisions through explainable AI.

Patient Portal → Doctor Portal

Trusted by leading healthcare institutions • Explainable AI • Clinical Decision Support

Diabetes Intelligence
Live AI clinical analysis
• Live

Glucose 112 mg/dL

Explainable AI
Top factors surfaced

HbA1c
6.7%

Risk Score
Medium

Diabetes Stage
Pre-Diabetes
Stage 2

Confidence
92%

AI Recommendation
Lifestyle intervention with glucose monitoring. Reassess HbA1c in 3 ...

Clinician Verified
Clinical trust

👋 Hi! I'm GlucoGuide AI. Ask me anything about diabetes or your prediction.

Conclusion

GlucoGuide

GlucoGuide integrates Machine Learning, OCR, RAG, and LLMs to provide:

- Diabetes prediction & risk assessment
- Automated medical report analysis
- Explainable AI-based insights
- Interactive clinical dashboard
- An integrated AI solution supporting accurate, explainable, and data-driven clinical decision-making.

Future Work

1

CGM Integration

Connect continuous glucose monitoring devices.

2

Clinical Integration

Connect the system with doctors and diabetes clinics.

3

Longitudinal Monitoring

Track patient risk and glucose trends over time.

4

RAG Expansion

Add multiple trusted clinical and nutritional references.

5

Advanced OCR

Support diverse laboratory report formats and improve extraction accuracy.

6

EHR Integration

Connect with electronic health records for seamless clinical data access.

7

Model Recalibration

Update and validate the model using new real-world clinical data.

8

Smart Alerts

Notify healthcare professionals of significant changes in patient risk.

Selected References

Key studies that shaped our approach and evaluation

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THANK YOU

السكر حكاية...
والوعي بداية.

لأن عادة صغيرة اليوم
ممكن تغيّر حياة بكرة.

GlucoGuide

From Prediction to Prevention